Department of Computer Science

Term Project Proposal, Fall 2024-25

Course	OBJECT ORIENTED PROGRAMMING 2	Group	

Group Members:

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Project Title:

Let's Travel

Project Description:

What is the problem against which you are planning to build the project? How does this solution help to solve the identified problem? (not more than 200 words)

When we plan for travel we face lots of problems like where will we go, how will we go, which hotel we will check in and where will we eat food. Most of time we just plan where we go then one by one we do other things. Let's Travel aims to simplify and enhance the process of planning and booking travel arrangements. Users can choose destinations, book transportation (bus, train, or flight), book hotels and reserve restaurants. The system integrates multiple services into one seamless platform, ensuring ease of use and efficiency for users. This solution provides a centralized and user-friendly interface, saving time and improving the overall travel planning experience.

Project Features:

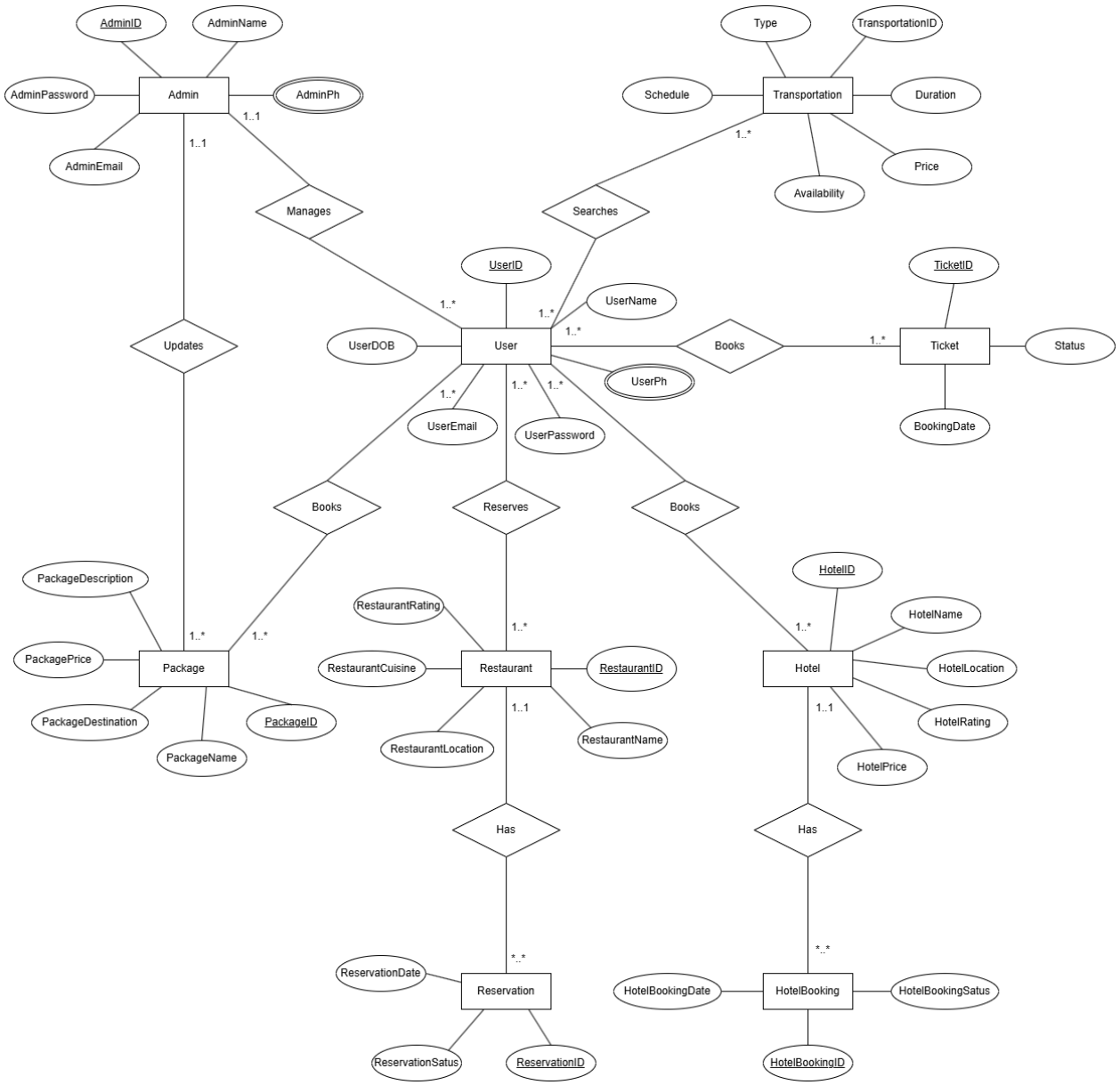
Admin:

- Login
- Update package
- Update ticket schedule
- Update hotel information
- Update restaurant information
- Add user
- Remove user

User:

- Login
- Registration
- Change password
- Choose package
- Search destination
- Choose how to go (bus, train, flight)
- Book tickets
- Cancel tickets
- Choose the hotel
- Book different class of hotel
- Book hotel room
- Cancel hotel booking
- Filter hotels by price
- Reserve the restaurant
- Filter restaurants by rating
- Cancel restaurants reservations
- Show the history
- Change the profile information

ER diagram:


AdminID
AdminName
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Manages
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ReservationDate
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ReservationSatus
ReservationID
HotelBookingDate
HotelBooking
HotelBookingSatus
HotelBookingID

Entities:

1. **Admin**
   - Attributes: AdminID (Primary Key), Username, Password, Email.
2. **User**
   - Attributes: UserID (Primary Key), Name, Email, Password, Phone, ProfileInfo, BookingHistory.
3. **Package**
   - Attributes: PackageID (Primary Key), Name, Destination, Price, Description.
4. **Transportation**
   - Attributes: TransportID (Primary Key), Type (Bus/Train/Flight), Schedule, Price, Duration, Availability.
5. **Ticket**
   - Attributes: TicketID (Primary Key), UserID (Foreign Key), TransportID (Foreign Key), BookingDate, Status (Booked/Canceled).
6. **Hotel**
   - Attributes: HotelID (Primary Key), Name, Location, Price, Class, Amenities, Rating.
7. **HotelBooking**
   - Attributes: BookingID (Primary Key), UserID (Foreign Key), HotelID (Foreign Key), BookingDate, Status (Booked/Canceled).
8. **Restaurant**
   - Attributes: RestaurantID (Primary Key), Name, Location, Cuisine, PriceRange, Rating.
9. **RestaurantReservation**
   - Attributes: ReservationID (Primary Key), UserID (Foreign Key), RestaurantID (Foreign Key), ReservationDate, Status (Reserved/Canceled).

Relationships:

1. **Admin-Manages-User**: One Admin can manage multiple Users.
2. **Admin-Updates-Package**: Admin can update multiple Packages.
3. **User-Books-Package**: One User can book multiple Packages.
4. **User-Searches-Transportation**: Users can search Transportation options.
5. **User-Books-Ticket**: Users book Tickets linked to a specific Transportation option.
6. **User-Books-Hotel**: Users can book multiple Hotels.
7. **User-Reserves-Restaurant**: Users can reserve tables at multiple Restaurants.
8. **Hotel-Has-Bookings**: One Hotel can have multiple bookings by different Users.
9. **Restaurant-Has-Reservations**: Restaurants can have reservations from multiple Users.

Example for User Stories:

Use Feature Name	Actor	Feature Story
login	admin	As a admin of the system, I want to login to the system to access the features of the system.
Update package	admin	Admin can update travel package in system
Update ticket schedule	admin	Admin can update ticket schedule in system
Update hotel information	admin	Admin can update hotel information in system
Update restaurant information	admin	Admin can update restaurant information in system
Add user	admin	Admin can add user in system
Remove user	admin	Admin can remove user in system
Login	user	User are login for access feature
Registration	user	User are registration for login and access the feature
Change password	user	User can change his password if he want or forget password
Choose package	user	User can choose package form given package he want to go
Search destination	user	User can search destination where he want to go
Choose how to go (bus, train, flight)	user	User can choose transport for travel like bus, train, flight
Book tickets	user	User can book the ticket of bus/train/flight

Cancel tickets	user	User can cancel travel transport which he book
Choose the hotel	user	User can choose hotel where he want to check in
Book different class of hotel	user	User can choose hotel of different class like 5 star 4 star 3 star
Book hotel room	user	User can choose different type of room
Cancel hotel booking	user	User can cancel hotel which he were book
Filter hotels by price	user	User also filter the hotel list by hotel rating
Reserve the restaurant	user	User can also reserve the restaurant where he want to eat
Filter restaurants by rating	user	User can also filter the restaurant list by restaurant rating
Cancel restaurants reservations	user	User also cancel the reservation
Show the history	user	User can see his travel booking history
Change the profile information	user	User also update his profile information